

QTag — JazzCash Topup API

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Audience: JazzCash integration team · **Owner:** iTechnologi / Malir Expressway (QTag)

This document describes the two HTTP endpoints QTag exposes for the JazzCash tag-topup flow:

1. **Tag Inquiry** — JazzCash looks up a tag (by chip **TID**) and shows the consumer their details before payment.
2. **Payment Notification** — after the customer pays in the JazzCash app, JazzCash notifies QTag, which credits the tag's balance and records the transaction.

1. Flow overview

```
Customer enters TID in JazzCash app
  |
  v
(1) POST /payments/jazzcash/inquiry/    -> QTag returns consumer details
  |
  Customer reviews + pays in JazzCash app
  |
  v
(2) POST /payments/jazzcash/payment/    -> QTag credits balance, returns new balance
```

- (1) is **read-only** (no state change).
- (2) is the **only** state-changing call and is **idempotent** (safe to retry).

2. Base URL & environment

Environment	Base URL
Production	<code>https://<qtag-host>/api/v1</code> (to be provided)
Test (LAN)	<code>http://<server-ip>:8000/api/v1</code>

All endpoint paths below are relative to the base URL.

Production must use HTTPS. The test environment runs on a LAN over HTTP.

3. Conventions

- **Method/Body:** `POST` with `Content-Type: application/json` . UTF-8.
- **Response envelope:** every response is JSON in this shape:

```
{ "success": true, "message": "OK", "data": { ... } }
{ "success": false, "message": "Tag not found", "errors": { ... } }
```

- `success` (boolean) — overall result. Check this first.
- `message` (string) — human-readable status / error.
- `data` (object) — present on success.
- `errors` (object, optional) — present on validation errors.

HTTP status codes:

Code	Meaning
200	OK (inquiry found / payment processed / duplicate ignored)
400	Bad request (missing/invalid field, amount ≤ 0)
401	Invalid signature (<code>pp_SecureHash</code> check failed)
404	Tag not found / not registered
500	Server error

4. Authentication — `pp_SecureHash`

Each request **should** include a `pp_SecureHash` for integrity/authentication.

Algorithm (HMAC-SHA256, uppercase hex):

1. Take all request fields **except** `pp_SecureHash` .
2. Drop empty values; sort the remaining by key (ascending).
3. Build the string: `SALT&key1=value1&key2=value2&...`
4. `HMAC_SHA256( key = SALT, message = string )` → hex → **UPPERCASE**.

```
pp_SecureHash = UPPER( HMAC_SHA256( SALT,
  "SALT&amount=500&jazzcash_txn_id=JC123&tid=E2801105..." ) )
```

- `SALT` = the shared **integrity salt** exchanged out-of-band.
- On QTag, enforcement is controlled by a server flag. During initial integration it may be **disabled** (any request accepted); for production it is **enabled** and unsigned/invalid requests get **401**.

Action item: confirm the exact hashing scheme JazzCash uses for `pp_SecureHash` . If it differs, QTag will align to JazzCash's documented standard before go-live.

5. Endpoint 1 — Tag Inquiry

Look up a tag by chip TID and return the consumer's details. Read-only.

POST /payments/jazzcash/inquiry/

Request

Field	Type	Required	Description
tid	string	yes	RFID chip TID (24 hex chars). Normalized server-side (spaces removed, uppercased).
pp_SecureHash	string	prod	Integrity hash (see §4).

```
{ "tid": "E28011052000704A8F9F0AE3", "pp_SecureHash": "9F3C..." }
```

Success — 200

```
{
  "success": true,
  "message": "OK",
  "data": {
    "consumer_id": "4f754655-5b4a-4f32-a7d2-0d508533822a",
    "vehicle_registration": "BP-3641",
    "customer_name": "Kashif Mughal",
    "tid": "E28011052000704A8F9F0AE3",
    "current_balance": "1500.00",
    "status": "active"
  }
}
```

data field	Type	Description
consumer_id	string (UUID)	QTag's stable consumer identifier (owner UUID).
vehicle_registration	string	Vehicle plate number.
customer_name	string	Registered consumer name.
tid	string	Echoed (normalized) TID.
current_balance	string (decimal)	Current wallet balance, in PKR .
status	string	active .

Not found / not registered — 404

```
{ "success": false, "message": "Tag not found" }
```

Other reasons: Tag not assigned to any vehicle, Tag is <status>, No account found for this vehicle.

Invalid signature — 401

```
{ "success": false, "message": "Invalid signature" }
```

6. Endpoint 2 — Payment Notification

Credit the tag's balance after the customer has paid. **Idempotent** on `jazzcash_txn_id`.

```
POST /payments/jazzcash/payment/
```

Request

Field	Type	Required	Description
<code>tid</code>	string	yes	RFID chip TID.
<code>amount</code>	string number	yes	Amount to credit, in PKR (e.g. <code>"500.00"</code>). Must be > 0. (unit: see action item)
<code>jazzcash_txn_id</code>	string	yes	JazzCash's unique transaction id. Used as the idempotency key . (<code>pp_TxnRefNo</code> also accepted.)
<code>pp_SecureHash</code>	string	prod	Integrity hash (see §4).

```
{
  "tid": "E28011052000704A8F9F0AE3",
  "amount": "500.00",
  "jazzcash_txn_id": "JC1234567890",
  "pp_SecureHash": "A7B1..."
}
```

Success — 200

```
{
  "success": true,
  "message": "Balance updated",
  "data": {
    "jazzcash_txn_id": "JC1234567890",
    "new_balance": "2000.00",
    "topup_id": "36ac6d87-7940-4425-8726-ed849dc9c97b"
  }
}
```

data field	Type	Description
jazzcash_txn_id	string	Echoed transaction id.
new_balance	string (decimal)	Balance after crediting, PKR.
topup_id	string (UUID)	QTag's topup record id.

Duplicate (already processed) — 200

A repeated jazzcash_txn_id does **not** credit again; QTag returns success with already_processed: true and the current balance:

```
{
  "success": true,
  "message": "Balance updated",
  "data": {
    "already_processed": true,
    "jazzcash_txn_id": "JC1234567890",
    "new_balance": "2000.00"
  }
}
```

Errors — 400 / 401 / 404

```
{ "success": false, "message": "Amount must be greater than zero" }
```

Other reasons: tid is required, jazzcash_txn_id is required, Invalid amount, Tag not found, Tag not assigned to any vehicle, No account found for this vehicle, Invalid signature (401).

7. Idempotency (important)

- jazzcash_txn_id is **unique** in QTag. If JazzCash retries the payment call (timeout, network), QTag detects the duplicate and returns 200 with already_processed: true — the balance is credited **exactly once**.
- Recommendation:** JazzCash should retry the payment call until it receives a 200 with success: true . Retries are safe.

8. Error catalogue

message	HTTP	Endpoint	Cause
Invalid signature	401	both	pp_SecureHash invalid (when enforced)
tid is required	400	both	Missing tid
Tag not found	404 / 400	inquiry / payment	TID not registered
Tag not assigned to any vehicle	404 / 400	both	Tag exists but unassigned
Tag is <status>	404	inquiry	Tag not active
No account found for this vehicle	404 / 400	both	No wallet for the tag
jazzcash_txn_id is required	400	payment	Missing txn id
Invalid amount	400	payment	Non-numeric amount
Amount must be greater than zero	400	payment	amount $\leq$ 0

9. Action items to confirm with JazzCash

1. **pp_SecureHash algorithm** — confirm JazzCash's exact scheme; QTag will match it. Exchange the shared SALT securely.
2. **Amount unit** — PKR rupees (e.g. 500.00) or paisa (50000)? Plus any **min/max** topup limits.
3. **Field names** — confirm JazzCash's request field names (tid , amount , jazzcash_txn_id) or provide their standard; QTag can map.
4. **Response codes** — does JazzCash expect a specific responseCode (e.g. 000) in the body? QTag can add a mapping.
5. **Source IPs** — provide JazzCash server IP range for an **allowlist**.
6. **HTTPS** — production endpoint will be HTTPS; confirm TLS requirements.

10. Changelog

Version	Date	Notes
v1 (draft)	2026-06-19	Initial inquiry + payment endpoints.